

X Lecture

Tests on a Covariance Matrix

10.1. Test of $\Sigma = \Sigma_0$.

We start with this simpler case since ideas are more transparent. The practically more relevant cases are about comparing covariance matrices of two or more multivariate normal populations but the derivations of the latter tests is more subtle. For these we will only formulate the final results.

Assume now that we have the sample $X_1, X_2, \dots, X_n$ from a $N_p(\mu, \Sigma)$ distribution and we would like to test $H_0 : \Sigma = \Sigma_0$ against the alternative $H_0 : \Sigma \neq \Sigma_0$. Obviously the problem can be easily transformed into testing $\bar{H}_0 : \Sigma = I_p$ since otherwise we can consider the modified observations $Y_i = \Sigma_0^{-\frac{1}{2}} X_i$ which under H_0 will be multivariate normal with a covariance matrix being equal to I_p . Therefore we can assume that:

$X_1, X_2, \dots, X_n$ is a sample from a $N_p(\mu, \Sigma)$ and we want to test $H_0 : \Sigma = I_p$ versus $H_1 : \Sigma \neq I_p$. We will derive the likelihood ratio test for this problem. The likelihood function is

$$L(\mathbf{x}; \mu, \Sigma) = (2\pi)^{-\frac{np}{2}} |\Sigma|^{-\frac{n}{2}} e^{-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \mu)' \Sigma^{-1} (\mathbf{x}_i - \mu)} = \\ (2\pi)^{-\frac{np}{2}} |\Sigma|^{-\frac{n}{2}} e^{-\frac{1}{2} \text{tr}[\Sigma^{-1} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)']}$$

Under the hypothesis H_0 , the maximum of the likelihood function is obtained when $\bar{\mu} = \bar{x}$. Under the alternative we have to maximize with respect to both μ and Σ and we know from Lecture III that the maximum of the likelihood function is obtained for $\hat{\mu} = \bar{x}$ and $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$. Then we obtain easily the likelihood ratio

$$\Lambda = \frac{\max_{\mu} L(x; \mu, I_p)}{\max_{\mu, \Sigma} L(x; \mu, \Sigma)} = \frac{e^{[-\frac{1}{2} \text{tr} V]}}{|V|^{-\frac{n}{2}} n^{\frac{np}{2}} e^{-\frac{np}{2}}}$$

where $V = \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$. Therefore :

$$-2 \log \Lambda = np \log n - n \log |V| + \text{tr} V - np \quad (10.1)$$

and according to the asymptotic theory the quantity in (10.1) is asymptotically distributed as $\chi_{p(p+1)/2}^2$ (the degrees of freedom being the difference of the number of free parameters under the alternative and under the hypothesis). This test would reject H_0 if the value of the $-2 \log \Lambda$ statistic is significantly large.

10.2. Sphericity test

Further, it is more realistic to assume that the structure of the covariance matrix is only known up to some constant. Having in mind the discussion in the beginning of **10.1**, we can assume without loss of generality that $H_0 : \Sigma = \sigma^2 I_p$ against a general alternative. This test has the name "sphericity test". The likelihood ratio test can be developed in a manner similar to the previous case (do it (!)) and the final result is that

$$-2 \log \Lambda = np \log(n\hat{\sigma}^2) - n \log |V|$$

Hereby $\hat{\sigma}^2 = \frac{1}{np} \sum_{i=1}^n (x_i - \bar{x})'(x_i - \bar{x})$. The asymptotic distribution of $np \log(n\hat{\sigma}^2) - n \log |V|$ under the null hypothesis will be again χ^2 but the degrees of freedom are this time $\frac{p(p+1)}{2} - 1 = \frac{(p-1)(p+2)}{2}$ (WHY (!)). Again, the hypothesis will be rejected for large values.

10.3. General situation. Testing equality of covariance matrices from k different multivariate normal populations $N_p(\mu_i, \Sigma_i), i = 1, 2, \dots, k$ is a very important problem especially in discriminant analysis and multivariate analysis of variance. The testing procedure based on the likelihood ratio testing has been implemented in SAS. Denote by:

k - number of populations; p - dimension of vector

n - total sample size $n = n_1 + n_2 + \dots + n_k$, n_i being the sample size for each population

$N = n - k; N_i = n_i - 1$

$$V = \frac{\prod_{i=1}^k |\text{Within SS Matrix}(i)|^{\frac{N_i}{2}}}{|\text{Pooled SS Matrix}|^{\frac{N}{2}}}, \rho = 1 - \left[\sum_{i=1}^k \frac{1}{N_i} - \frac{1}{N} \right] \frac{2p^2 + 3p - 1}{6(p+1)(k-1)}$$

then under the null hypothesis of equality of all k covariance matrices, it holds:

$$-2\rho \log \frac{N^{pN/2} V}{\prod_{i=1}^k N_i^{pN_i/2}} \quad (10.2)$$

is asymptotically distributed as $\chi^2_{\frac{1}{2}(k-1)p(p+1)}$. Large values of the statistic are significant and lead to the rejection of the hypothesis about equality of the k covariance matrices.

We would like to put the result (10.2) into context. We will avoid the subtle details and refer to Chapter 8 of the monograph

Muirhead, R. (1982) Aspects of Multivariate Statistical Theory. Wiley, New York.

First, it is easy to see that when N_i and N are replaced by n_i and n respectively and ρ is set to 1 then (10.2) is nothing else then the *classical* Likelihood Ratio (LR) test for equality of the covariance matrices. The LR test is precisely the one that is suggested as the “statistic” $\eta = -2 \ln \lambda_2$ on p. 233 in the hard copy given to you during the Discriminant analysis lab.

It has been noticed that this test has the defect that it is (asymptotically) biased: that is, the probability of rejecting H_0 when H_0 is false can be smaller than the probability of rejecting H_0 when H_0 is true (i.e., it may happen that in some points of the parameter space the probability of a correct decision is smaller than the probability for a wrong decision). Hence it is desirable to modify it to make it asymptotically unbiased. This *modified* LR is achieved precisely by replacing n_i and n by N_i and N (that is, by the correct degrees of freedom). In this way (and if $\rho = 1$) we would get (10.2).

We note that indeed $\rho = 1 - \left[\sum_{i=1}^k \frac{1}{N_i} - \frac{1}{N} \right] \frac{2p^2 + 3p - 1}{6(p+1)(k-1)}$ is close to 1 anyway if all sample sizes n_i were very large. Finally, the replacement of $\rho = 1$ by $\rho = 1 - \left[\sum_{i=1}^k \frac{1}{N_i} - \frac{1}{N} \right] \frac{2p^2 + 3p - 1}{6(p+1)(k-1)}$ that is made in (10.2) serves to improve the quality of the asymptotic approximation of the statistic by the limiting $\chi^2_{\frac{1}{2}(k-1)p(p+1)}$ distribution. Such (asymptotically negligible) scalar transformations of the LR statistic that yield improved test statistic with a chi-squared null distribution of order $O(1/n)$ instead of the ordinary $O(1)$ for the standard LR, are known in the literature under the common name **Bartlett corrections**. Thus (10.2) is a Bartlett corrected version of the modified LR statistic. The statistic (10.2) is the one that is implemented in SAS.